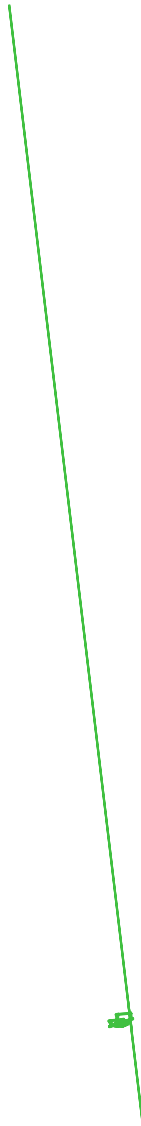


AeroAssembly_assy_example/AeroAssembly_conne



Stage 3, connect. The same eight parts with no 'location:' anywhere: each names the item it is joined to, the M3 pattern or the seat it is joined by, and what somebody has to do to make that joint.

Package: //pub/examples/partcad/feature_import

Assemblies: 1

Steps: 7

Generated by PartCAD

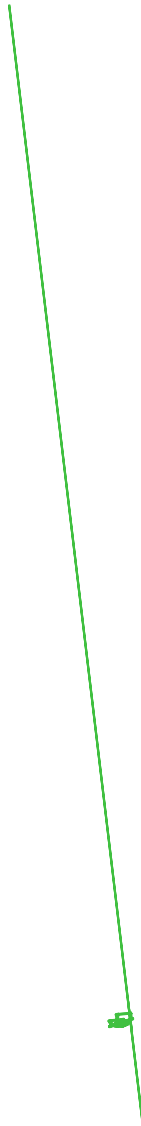
Bill of Materials

Parts

[//pub/examples/partcad/feature_import](#)

Part	Count	Description
AeroAssembly_assy_example/AeroFrameAssembly_solid1	1	The first frame half's rear foot
AeroAssembly_assy_example/AeroFrameAssembly_solid2	1	The first frame half's front foot
AeroAssembly_assy_example/AeroFrameAssembly_solid3	1	The first frame half's strut
AeroAssembly_assy_example/AeroFrame_Cap	1	The cap, which stands on the plate on four feet - two of which touch
AeroAssembly_assy_example/AeroFrame_Plate	1	The plate both frame halves bolt to and the cap stands on
AeroAssembly_assy_example/MirrorAeroFrameAssembly_solid1	1	The mirrored half's rear foot
AeroAssembly_assy_example/MirrorAeroFrameAssembly_solid2	1	The mirrored half's front foot
AeroAssembly_assy_example/MirrorAeroFrameAssembly_solid3	1	The mirrored half's strut

AeroAssembly_assy_example/AeroAssembly_conne



Stage 3, connect. The same eight parts with no 'location:' anywhere: each names the item it is joined to, the M3 pattern or the seat it is joined by, and what somebody has to do to make that joint.

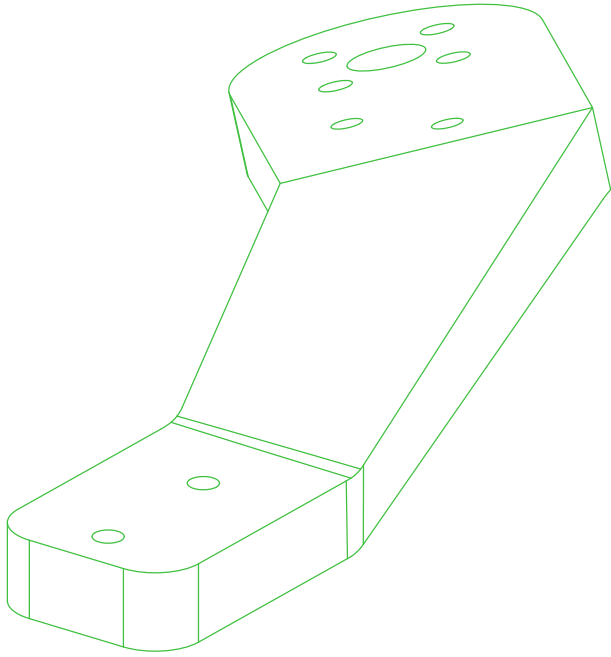
Package: //pub/examples/partcad/feature_import

Steps: 7

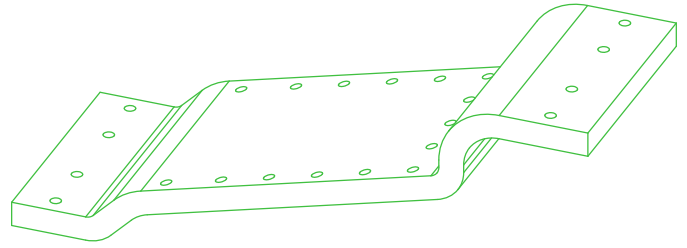
Start with AeroFrame_Plate.

The plate: two flanges, eight Ø3 clearance holes, and the top face the cap stands on.

AeroAssembly_assy_example/AeroAssembly_conne



AeroFrameAssembly_solid2

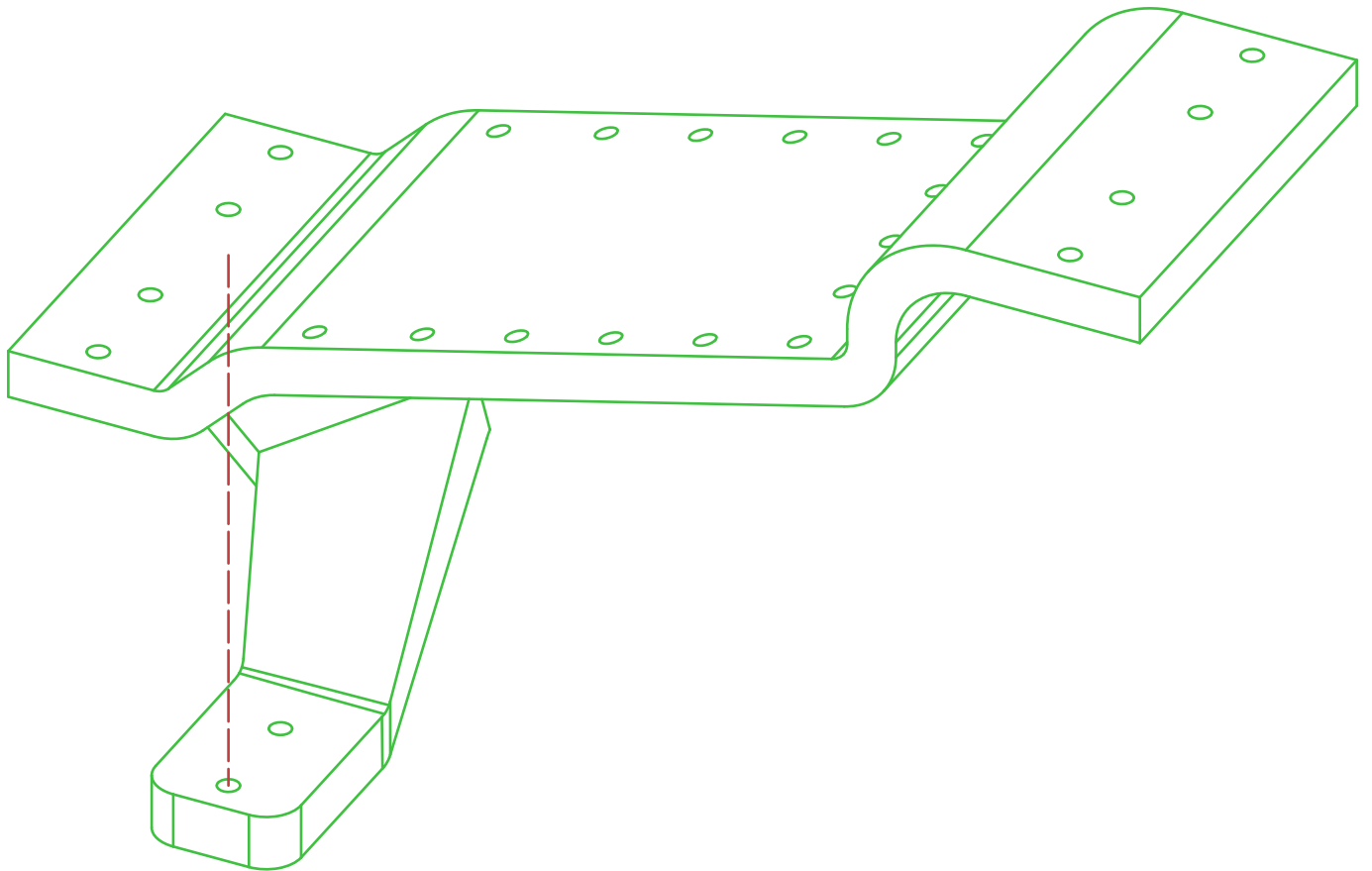


AeroFrame_Plate

The front foot of the first frame half

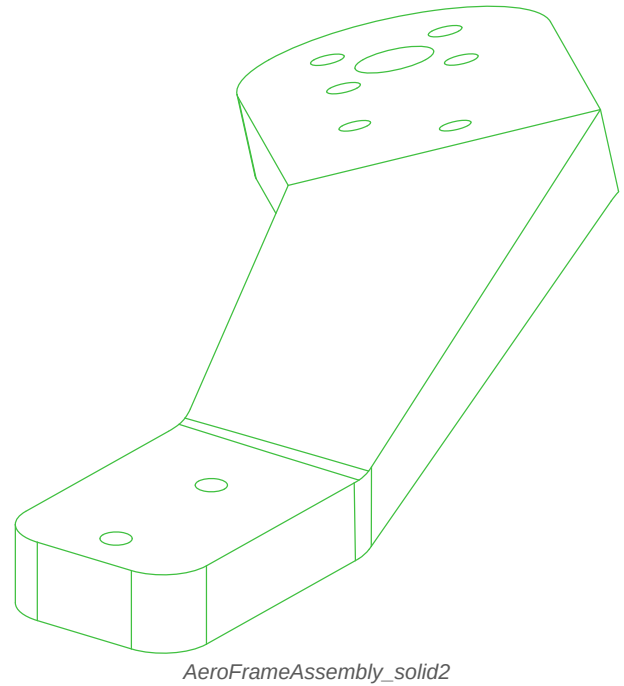
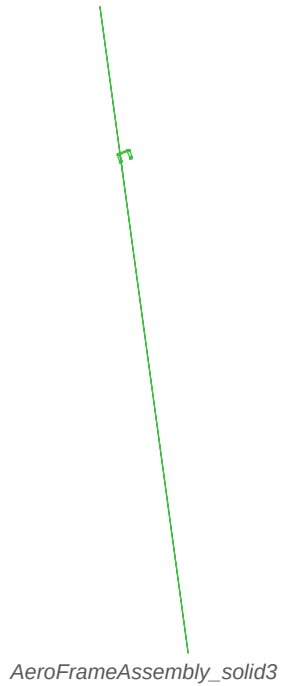
Connect the plate-a-8mm-thru-opening-m3 port of AeroFrameAssembly_solid2 to the frame-front-a-7mm-thru-opening-m3 port of AeroFrame_Plate.

Note: Offer the foot up under the plate's front flange and start one bolt - the 'a' hole of the pair - before the other, so the foot can still swing onto the second. Both holes are ISO 273 medium clearance for an M3, 3.4mm, so neither threads anything: what holds the joint is the bolt and its nut, which this assembly does not model.



Exploded view: the two are shown 81.2mm apart.

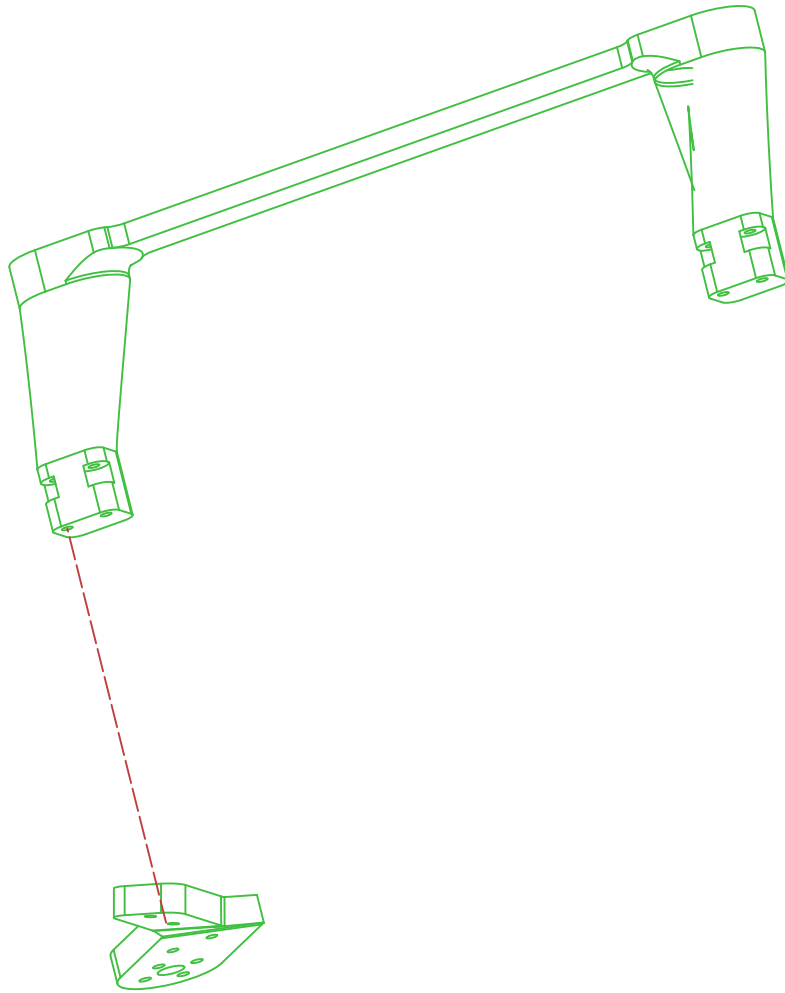
AeroAssembly_assy_example/AeroAssembly_conne



The strut of the first frame half

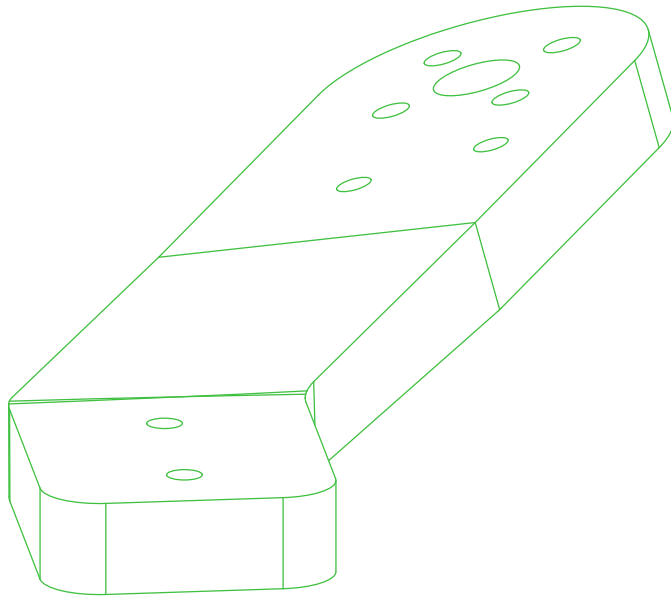
Connect the foot-a-6mm-thru-opening-m3 port of AeroFrameAssembly_solid3 to the strut-a-8mm-thru-opening-m3 port of AeroFrameAssembly_solid2.

Note: The strut seats flat on the foot and is bolted to it through the flange 14mm above that seat, so the bolt is long enough to cross the channel. Start both bolts before tightening either: the strut is what squares the half up, and it can only do that while it is free to move.



Exploded view: the two are shown 113.9mm apart.

AeroAssembly_assy_example/AeroAssembly_conne



AeroFrameAssembly_solid1

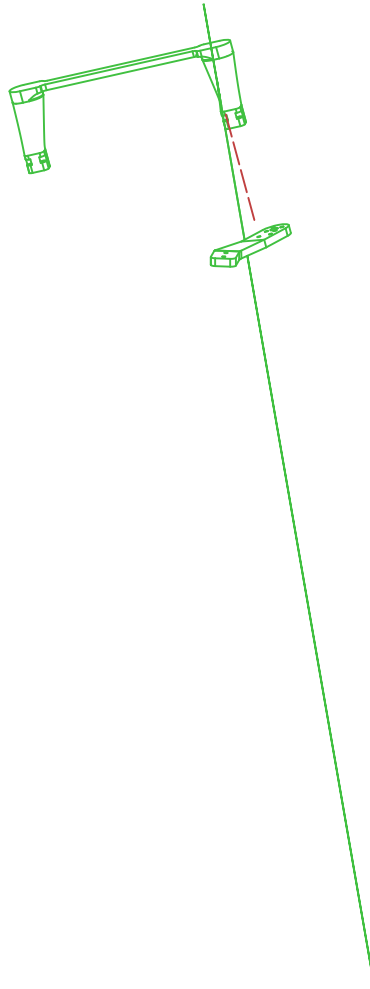


AeroFrameAssembly_solid3

The rear foot of the first frame half

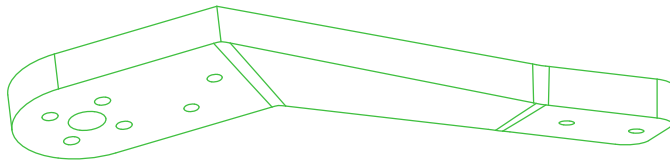
Connect the strut-a-8mm-thru-opening-m3 port of AeroFrameAssembly_solid1 to the rear-a-6mm-thru-opening-m3 port of AeroFrameAssembly_solid3.

Note: The rear foot goes on the far end of the strut, and the two bolts that hold it are the last of this half. It then bears on the underside of the plate's rear flange - bears, and nothing more: the flange's four $\varnothing 3$ holes are 2.5mm away from this foot's two, in the source file and therefore in all three stages, so there is nothing to bolt there. Both patterns are declared on the parts; neither is a joint.

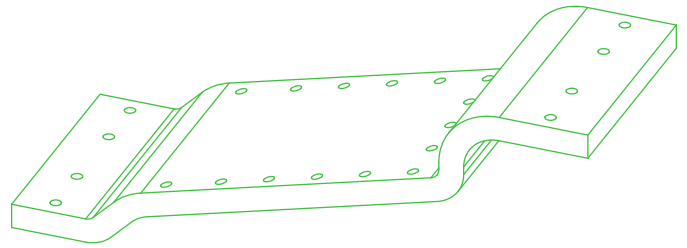


Exploded view: the two are shown 113.9mm apart.

AeroAssembly_assy_example/AeroAssembly_conne



MirrorAeroFrameAssembly_solid2

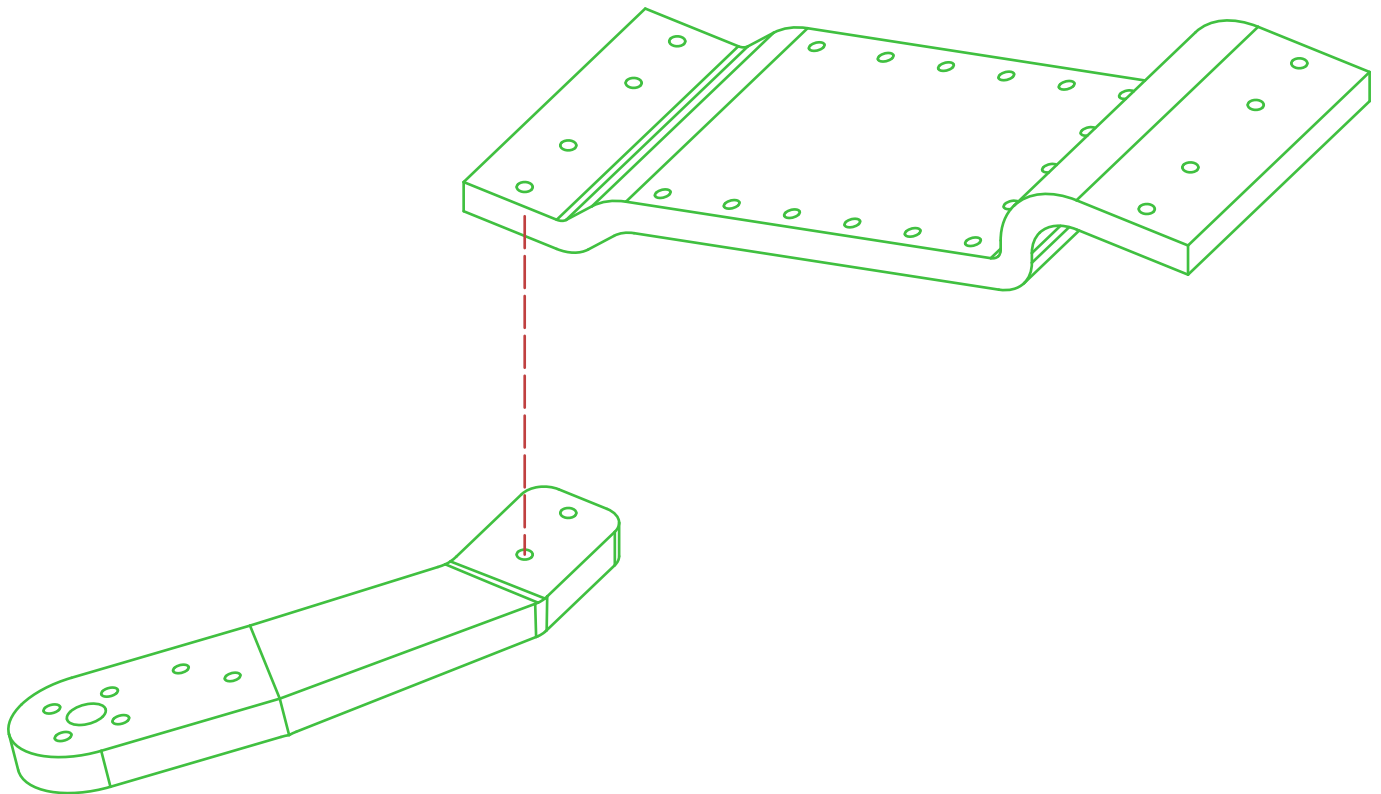


AeroFrame_Plate

The front foot of the mirrored frame half

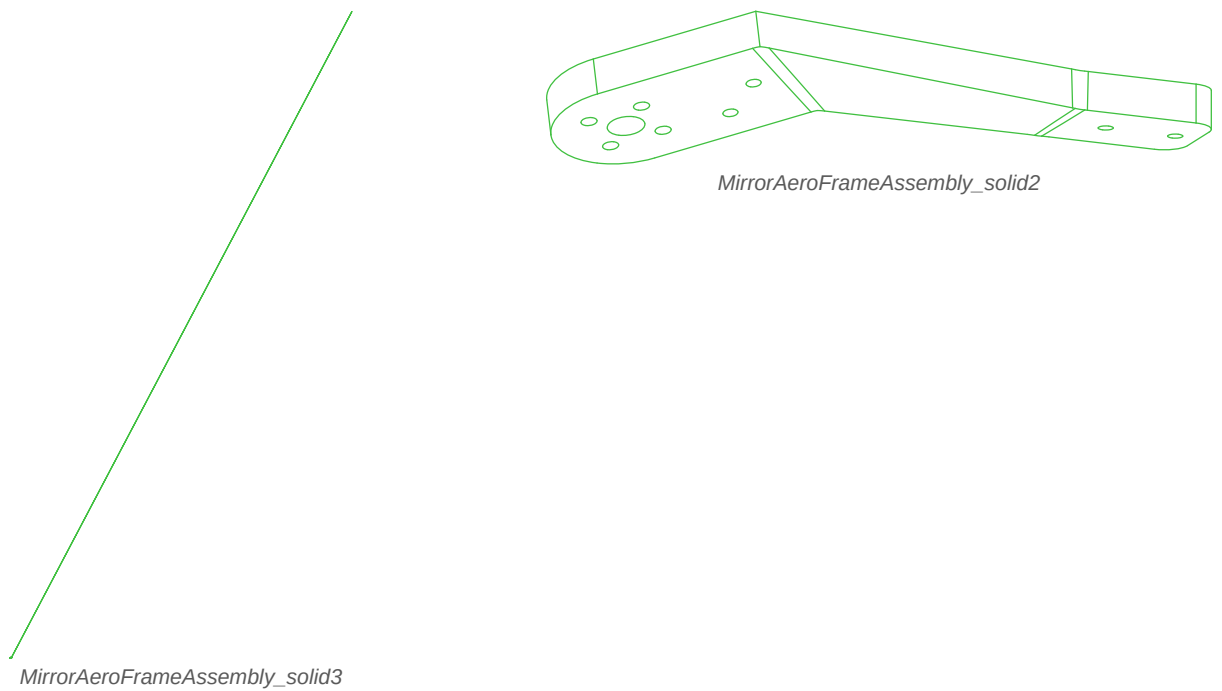
Connect the plate-a-8mm-thru-opening-m3 port of MirrorAeroFrameAssembly_solid2 to the mirror-front-a-7mm-thru-opening-m3 port of AeroFrame_Plate.

Note: The other pair of holes in the same flange. In the import this foot was (7.728713, -3.380867, 2.583602) mm away from them and its own half was 2.583602 mm clear of the flange at one end and 12.997 mm³ inside the plate at the other. Connected rather than placed, it cannot be: the holes are where it goes.



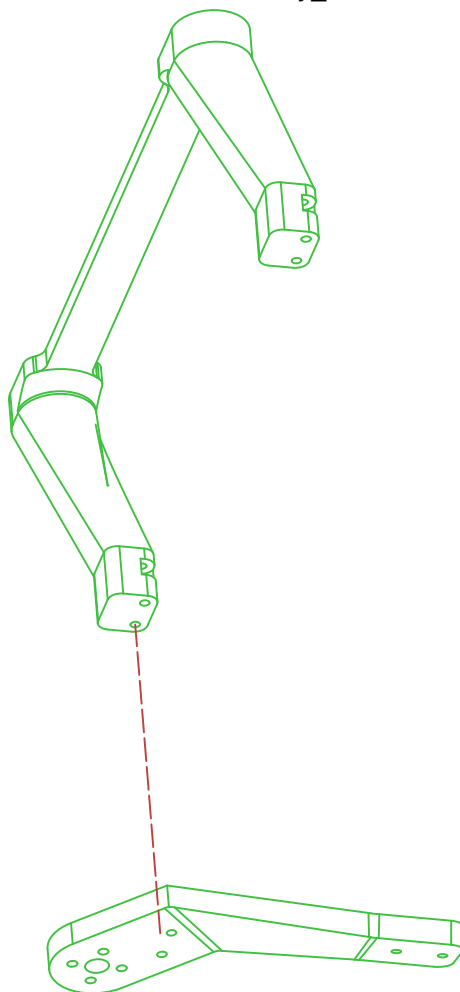
Exploded view: the two are shown 81.2mm apart.

AeroAssembly_assy_example/AeroAssembly_conne



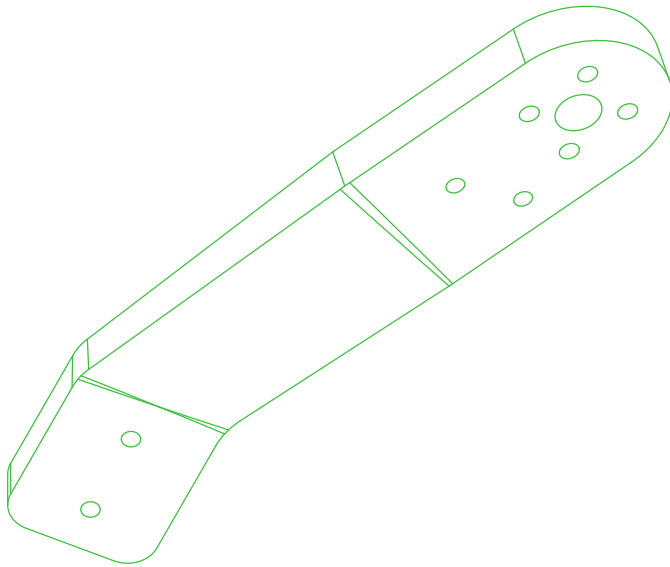
The strut of the mirrored frame half

Connect the foot-a-6mm-thru-opening-m3 port of *MirrorAeroFrameAssembly_solid3* to the strut-a-8mm-thru-opening-m3 port of *MirrorAeroFrameAssembly_solid2*.

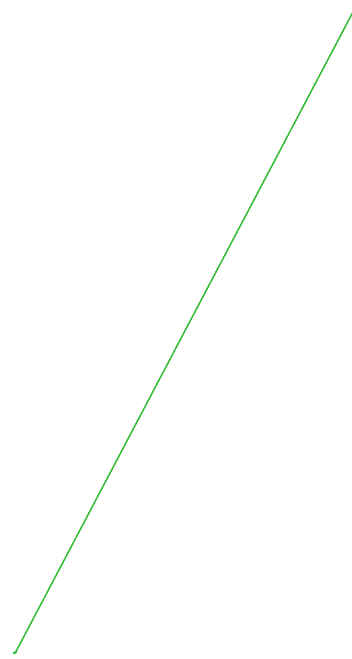


Exploded view: the two are shown 117.5mm apart.

AeroAssembly_assy_example/AeroAssembly_conne



MirrorAeroFrameAssembly_solid1

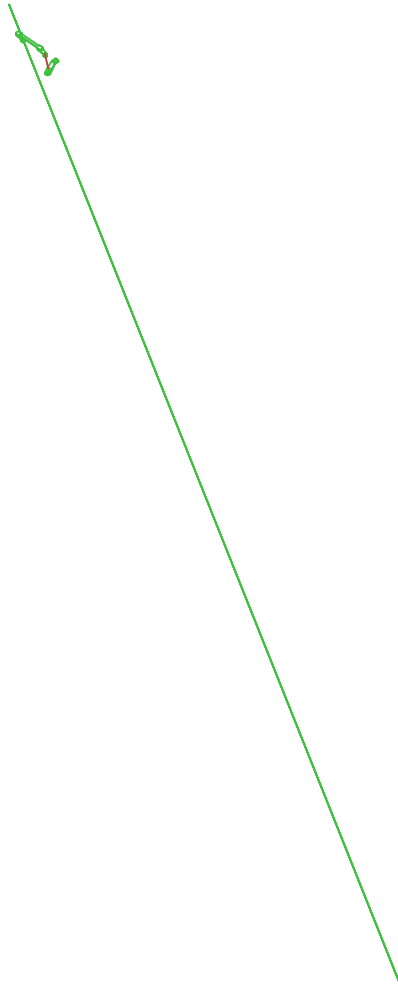


MirrorAeroFrameAssembly_solid3

The rear foot of the mirrored frame half

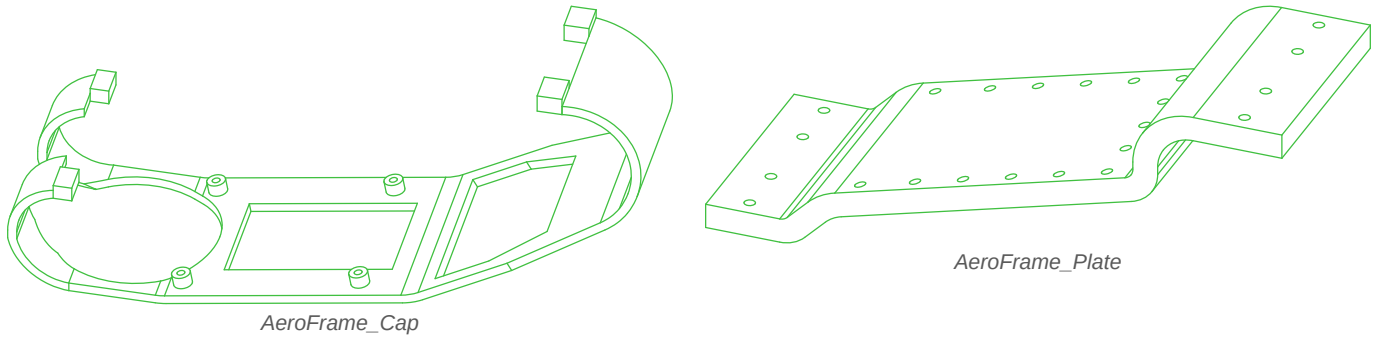
Connect the strut-a-8mm-thru-opening-m3 port of MirrorAeroFrameAssembly_solid1 to the rear-a-6mm-thru-opening-m3 port of MirrorAeroFrameAssembly_solid3.

Note: The mirror image of the first half's rear foot, down to its two holes being 2.5mm off the flange's four - the other way, which is what says the mismatch is in the design rather than in one instance of it.



Exploded view: the two are shown 117.5mm apart.

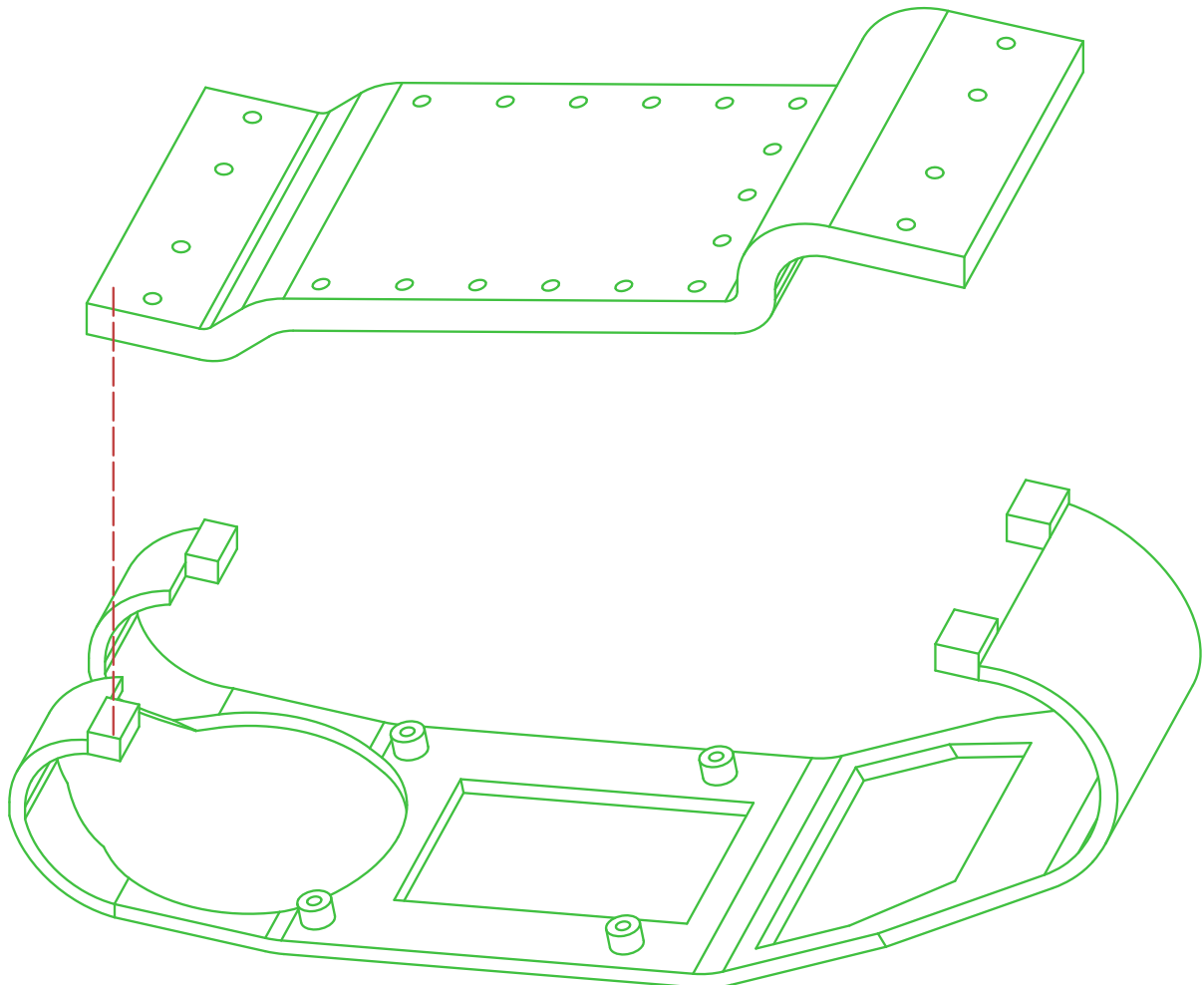
AeroAssembly_assy_example/AeroAssembly_conne



The cap, which stands on the plate

Connect the plate-a-seat port of *AeroFrame_Cap* to the cap-a-seat port of *AeroFrame_Plate*.

Note: The cap is not fastened: it stands on the plate on four feet, and only two of them touch. Its front feet are modelled 0.751923 mm too long for the faces they land on - the defect stage 1 reports as 90.231 mm³ of metal inside metal - so what is stated here is the pair that does seat, and the rear pair stands 0.751923 mm clear. Shim it there; nothing fastens through it.



Exploded view: the two are shown 102.5mm apart.

Links

This assembly: [AeroAssembly_assy_example/AeroAssembly_connected](#) *AeroAssembly_connected.assy*

This package: [//pub/examples/partcad/feature_import](#) *../README.md*

PartCAD <https://partcad.org/>